

PIT STOP CHALLENGE

Motorsport Tyre Change Competition

Mecha Rush'26 · Dept. of Mechanical Engineering · BSACIST

EVENT BRIEF

A motorsport-inspired technical event where teams race against the clock to complete a full tyre swap on a Provided vehicle; using only the tools provided, within a marked zone, and under real competition conditions.

Speed matters, but it is not everything. Proper jacking, accurate tyre fitment, secure fastening, and disciplined tool management all factor into the outcome.

VENUE & TENTATIVE SCHEDULE

Venue: Mechanical Science Block, 1st Floor - Front Side | April 7, 2026

Time	Activity
10:00 – 10:15 AM	Team verification & safety briefing
10:15 AM – 1:15 PM	Slot-wise competition
1:15 – 2:00 PM	Lunch
2:00 – 2:30 PM	Evaluation & result announcement
3:30 PM	Certificate distribution

TEAMS

- Exactly 4 members per team; no more, no fewer
- All 4 members must actively participate in the tyre change
- Cross-department teams permitted; cross-college teams not allowed
- One designated member operates the buzzer to start and stop the timer
- All participants must carry their college ID and conformation mail

SAFETY REQUIREMENTS

- Closed-toe shoes are mandatory for all participants — no exceptions
- All safety instructions from event coordinators must be followed at all times
- The jack, wheels, and tools must be handled carefully throughout the run
- Unsafe behavior will result in a penalty or immediate disqualification at the judge's discretion

EVENT PROCEDURE

Each team follows this sequence during their timed run:

1. Our driver stepped out of the vehicle - the timer starts.
1. The team safely jacks the provided vehicle at the designated lift point.
2. All existing tyres are removed and replaced with the provided tyres.
3. Each tyre must be correctly aligned and all nuts fully tightened before the next step.
4. All tools are returned to the designated place; no tools may remain in the active zone.
5. Final time = run time + all accumulated penalties.

6. After completing all the tasks, one of the team members shows a Thumbs up - the timer stops.

All participants must remain within the marked working zone throughout the run.

PENALTY SYSTEM

Final Time = Run Time + Cumulative Penalties

Penalty	Offence
+5 sec	Careless tool handling or improper tool use without vehicle damage
+5 sec	Minor delay in returning tools to designated rack after use
+5 sec	Crossing the working boundary line unnecessarily
+10 sec	Incorrect tool selected for the task
+10 sec	One participant taking over roles assigned to other team members
+10 sec	Loose fastening of bolts or nuts after task completion
+10 sec	Delay in removing the jack after work is done
+20 sec	Repeated dropping of tools
+20 sec	Unsafe or improper handling of the jack
+20 sec	Interference with another team's setup or zone
+10 sec (Stop-and-Go)	Ignoring a judge's instruction during the run
+10 sec (Stop-and-Go)	Violation of safety rules during the pitstop

JUDGING CRITERIA

Each run is evaluated across five criteria.

Criterion	Description
Completion Time	Primary metric- raw time from start signal to buzzer press
Tyre Installation	Correct alignment and seating of all replacement tyres
Nut Tightening	All fasteners fully and securely tightened
Tool Handling	Correct selection, usage, and return of tools to designated rack
Team Coordination	Role distribution, communication, and workflow efficiency

DISQUALIFICATION

- Provided vehicle is damaged due to negligence
- Tools are damaged or broken through mishandling
- Use of unauthorized tools not provided by the organizers
- Team continues working after the judge signals stop
- Unsportsmanlike conduct toward judges or other teams
- Repeated or flagrant safety violations at judge's discretion

The organising committee has the right to modify rules, participants will get informed if changes made

Judges' decisions are final. Event coordinators reserve the right to modify rules for smooth execution.

For queries, contact the organizing committee through contact@mecharush.in | mecharush.in